

Chapter 05 Raw Textbook Content

Textbook of Minimally Invasive Spine Surgery (Springer Nature 2026) | Full Raw Text Extraction

Textbook of Minimally Invasive Spine Surgery, Concepts and Surgical Techniques

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5. Transforaminal Endoscopic Discectomy

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Lumbar disc herniation, characterized by the displacement of the intervertebral disc material causing compression of the adjacent nerve roots, is among the most prevalent degenerative pathologies of the lumbar spine. According to current clinical guidelines, surgical intervention via discectomy is indicated when conservative treatment has failed [1]. Open surgical discectomy involves disruption of the surrounding tissues, including muscles, blood vessels, and the nerves supplying these muscles. This can result in muscle denervation and weakening, potentially leading to decreased muscle strength and chronic pain [2]. Other complications associated with a laminectomy include epidural fibrosis and arachnoiditis. Additionally, postoperative complications related to spinal instability, such as post-laminectomy instability, may occur [3].

Minimally invasive procedures have been developed faster during

this decade. Endoscopic discectomy has experienced rapid growth in the spine community. The endoscopic techniques for addressing lumbar disc herniation are categorized into interlaminar, paraspinal, and transforaminal (TF) approaches, each offering its own set of advantages and disadvantages. The selection of the most suitable approach remains a subject of debate and should be determined based on the patient's anatomy, pathology, and the surgeon's expertise [4]. The transforaminal approach is widely recognized as the least invasive technique for performing lumbar discectomy. The approach through Kambin's triangle—an anatomical region situated between the exiting and traversing nerve roots—has become a widely accepted technique for the surgical removal of herniated nucleus pulposus (HNP). Recent advancements in this method have expanded its application to more complex spinal pathologies. For example, it can now be effectively combined with foraminoplasty to treat foraminal stenosis, and in certain cases, partial resection of the pedicle may be necessary to retrieve disc fragments that have migrated superiorly or inferiorly. Comparative studies have shown that endoscopic decompression offers favorable outcomes with a lower complication rate when compared to conventional open surgery [5, 6].

5.1 Anatomic Consideration in Transforaminal Endoscopic Discectomy

Kambin defined a triangular anatomical area known as the